APSP Phase 2: Model Implementation

We implemented the Allocation and Periodic Scheduling Problem (APSP) optimization models presented in the selected research paper using Python, DOcplex, and IBM CPLEX.

To validate the implementation, we reproduced the three-task example from the paper. We implemented the bounded Monolithic MILP and Benders decomposition, including basic and accelerated versions.

Result: All three approaches obtained the same optimal period reported in the paper: **μ = 5**. The resulting schedules passed checks for allocation, dependency, and resource constraints.

Both Benders versions converged to the optimal solution with a zero optimality gap. The scheduling subproblem and its dual also produced matching objective values, supporting the correctness of the implementation.